# Identifying Musical Style from Audio Features: A Case Study of Angine de Poitrine

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***Abstract*—This project investigates whether the distinctive musical style of Angine de Poitrine (AdP) can be identified from audio features alone. A dataset of 96 AdP excerpts and 100 GTZAN rock excerpts is analyzed using features related to rhythm, pitch/tonality, and timbre. Rhythmic irregularity and deviation from 12-tone equal temperament (12-TET) are tested directly, while logistic regression and Support Vector Machine (SVM) classifiers are evaluated using grouped cross-validation. The results do not support the hypotheses of greater rhythmic irregularity or microtonality in AdP, but rhythm and pitch features nevertheless distinguish AdP from GTZAN rock well above the majority-class baseline. Timbre provides the strongest overall separation.**

***Index Terms*—music information retrieval, audio feature extraction, music classification, rhythmic irregularity, microtonality, timbre, support vector machines**

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## I. Introduction

Music Information Retrieval (MIR) applies computational methods to extract and analyze information from musical signals. Audio recordings can be represented using numerical features related to musical properties such as rhythm, pitch, and timbre, which can subsequently be analyzed statistically or used for classification [1].

The music of Angine de Poitrine (AdP) provides an interesting case study because its unconventional style motivates hypotheses about rhythmic irregularity and microtonality. In this study, the rock category of the GTZAN Genre Collection is used as a more conventional rock comparison set [2]. Rather than assuming that these perceived stylistic differences are directly visible in audio features, the analysis tests whether they can be measured computationally.

The research question is:

> *Can the unusual musical style of AdP be identified from audio features alone, and which measurable characteristics, particularly rhythmic irregularity and microtonality, separate it from conventional rock?*

This question is investigated through three hypotheses:

- H1 (rhythm): AdP exhibits greater rhythmic irregularity than GTZAN rock.
- H2 (pitch): AdP deviates more from 12-tone equal temperament (12-TET).
- H3 (classification): Rhythm and pitch features separate AdP from GTZAN rock clearly above a majority-class baseline.

## II. Theoretical Background

### A. Music Information Retrieval

Music Information Retrieval (MIR) uses computational methods to represent and analyze musical signals. A common approach is to extract numerical audio features that describe relevant properties of a recording and use these features for tasks such as music classification [1], [2].

### B. Audio Feature Extraction

Audio features are commonly calculated over short, overlapping time frames so that changes within a recording can be represented. Spectral representations such as the short-time Fourier transform (STFT) provide the basis for many frequency- and pitch-related descriptors [1]. Frame-level feature sequences can then be summarized using statistics such as their mean and standard deviation to obtain fixed-length representations of audio excerpts.

### C. Rhythmic Features

Rhythmic features describe temporal properties such as onset timing, tempo, periodicity, and beat structure. An onset represents the detected beginning of a musical event, while the time between consecutive onsets is called an inter-onset interval (IOI) [1]. In this study, rhythmic irregularity is primarily represented by the coefficient of variation (CV) of the IOIs,

$$CV_{\mathrm{IOI}} = \frac{\sigma_{\mathrm{IOI}}}{\mu_{\mathrm{IOI}}}, \tag{1}$$

where a larger value indicates greater variation in onset spacing relative to the mean interval. Additional rhythm descriptors capture complementary information about tempo, pulse clarity, and rhythmic periodicity.

### D. Pitch and Tonal Features

The pitch representation is based on 12-tone equal temperament (12-TET), in which an octave is divided into twelve logarithmically spaced semitones, with one semitone corresponding to 100 cents [1]. Chroma features summarize spectral energy into twelve pitch classes and provide a compact representation of tonal content.

Microtonality is operationalized in this study as the distance in cents between an estimated fundamental frequency $f_0$ and the nearest 12-TET semitone. Reliable pitched frames are identified using probabilistic YIN (pYIN), while YIN provides continuous fundamental-frequency estimates [3], [4]. Fundamental-frequency estimation is challenging in polyphonic recordings because several sound sources and harmonics can overlap simultaneously [1].

### E. Timbral Features

Timbre describes perceptual differences in sound quality and is strongly related to spectral characteristics [1], [2]. Standard timbral descriptors include spectral properties and mel-frequency cepstral coefficients (MFCCs), which provide a compact representation of spectral shape and are commonly used in music classification [2].

### F. Classification Methods

Supervised classification learns a decision rule from labeled feature vectors. Logistic regression provides a linear probabilistic classifier, while a support vector machine (SVM) with a radial basis function (RBF) kernel can represent nonlinear decision boundaries [5]. Comparing the two models therefore provides a simple comparison between linear and nonlinear separation of the extracted audio features.

## III. Method

### A. Data Preparation

The dataset consists of 196 nearly balanced 30-second audio excerpts. The Angine de Poitrine (AdP) class contains 96 excerpts derived from twelve tracks from the albums *Vol. 1* and *Vol. 2*. The original recordings were purchased digitally and divided into non-overlapping 30-second segments. Eight segments, distributed across each track, were retained per song, resulting in $12 \times 8 = 96$ AdP excerpts.

The comparison class consists of 100 pre-cut 30-second excerpts from the rock category of the GTZAN Genre Collection [2]. The GTZAN audio files used in this project were downloaded from the Kaggle website [6].

All audio was loaded as mono at a sampling rate of 22,050 Hz. Before feature extraction, each clip was root-mean-square (RMS) normalized to a common target of $-20$ decibels relative to full scale (dBFS) to prevent overall loudness from acting as a simple classification cue. The source track of every excerpt was retained so that multiple excerpts from the same AdP song could be kept within the same cross-validation fold.

### B. Feature Extraction

A total of 76 audio features were extracted from each excerpt: 13 rhythm features, 25 pitch/tonal features, and 38 timbre features. Mel-frequency cepstral coefficients (MFCCs) and the custom short-time descriptors used 25 ms analysis windows with a 10 ms hop, while pitch-oriented analysis used approximately 100 ms frames to provide greater frequency resolution.

*1) Rhythm:* Rhythm descriptors include estimated tempo, tempo variability, onset rate, inter-onset interval statistics, pulse clarity, tempogram entropy, beat-synchronization deviation, and onset-envelope statistics [1]. The primary H1 feature is `rhy_ioi_cv`, the coefficient of variation of inter-onset intervals.

*2) Pitch and tonality:* Pitch features include tuning, chromagram statistics, chroma entropy, spectral-grid descriptors, and fundamental-frequency estimates. Probabilistic YIN (pYIN) and YIN were used to identify reliable pitched frames and estimate $f_0$ [3], [4]. The primary H2 feature, `pit_yin_dev_mean`, measures the mean absolute distance in cents from the nearest 12-tone equal temperament (12-TET) semitone after correcting for global tuning [1].

*3) Timbre:* Timbre is represented using zero-crossing statistics, spectral centroid, bandwidth, rolloff, flatness, frame-level RMS statistics, and 13 MFCCs [1], [2].

Frame-level descriptors are summarized using temporal statistics such as the mean and standard deviation to obtain fixed-length feature vectors.

### C. Hypothesis Analysis

H1 and H2 were evaluated directly before classification. H1 was tested using `rhy_ioi_cv`, while H2 was tested using `pit_yin_dev_mean`. Because several AdP excerpts originate from the same song, clip-level values were aggregated to source-track medians before statistical inference to avoid treating excerpts from the same track as independent observations.

Group differences were evaluated using a two-sided Mann–Whitney $U$ test with a significance level of $\alpha = 0.05$ [7]. Cliff's delta was used as a non-parametric effect-size measure [8]. H1 used 12 AdP and 100 GTZAN source groups, while reliable $f_0$ estimates for H2 were available for 11 AdP and 56 GTZAN source groups.

### D. Classification

*1) Rhythm and pitch classification (H3):* H3 evaluates whether rhythm and pitch features can distinguish AdP from GTZAN rock above a majority-class baseline. Six $f_0$-based features were excluded from classification because reliable $f_0$ estimates were unavailable for many clips; this prevents patterns of missing values from becoming an artificial classification cue. The retained classification set therefore contains 70 features: 13 rhythm, 19 pitch/tonal, and 38 timbre descriptors. H3 itself is evaluated using the 32 retained rhythm and pitch features, while the timbre and full feature sets are evaluated separately in the feature-ablation analysis.

Three models were evaluated: a majority-class dummy classifier, Logistic Regression, and a support vector machine (SVM) with a radial basis function (RBF) kernel. Logistic regression provides a linear classifier, while the RBF SVM allows nonlinear class boundaries [5]. The SVM used $C = 1.0$ and $\gamma = \texttt{scale}$.

Models were evaluated using five-fold stratified grouped cross-validation (`StratifiedGroupKFold`, seed 42). Grouping by source track ensured that all excerpts originating from a given AdP track remained within the same fold, while each GTZAN excerpt was treated as a separate source group. Median imputation and feature standardization were fitted only on the training portion of each fold.

Performance was evaluated using accuracy, precision, recall, and $F_1$-score [1]. A confusion matrix was additionally examined for the main rhythm-and-pitch SVM classifier.

*2) Feature ablation:* To determine which musical dimensions drive performance, models were evaluated across distinct feature subsets: rhythm alone (13 features), pitch/tonality alone (19 features), rhythm and pitch combined (32 features), timbre alone (38 features), and all retained features combined (70 features). Model performance was evaluated across folds using accuracy, precision, recall, and $F_1$-score [1], reported as the mean and standard deviation across the five validation folds.

*3) Recording-bandwidth confound check:* Because AdP tracks originate from modern digital releases in the 2020s, whereas the GTZAN dataset was created in 2001 [6], the approximately 20-year difference in recording, mastering, and audio encoding eras could introduce systematic bandwidth differences. To ensure that classifiers were not merely distinguishing between production eras rather than musical style, two checks were performed:

*a) Diagnostic check:* To test whether high-frequency content could serve as an unintended shortcut for classification, the proportion of spectral energy above 9 kHz relative to total signal energy was calculated for each clip as a dedicated diagnostic feature. Classifiers were then trained and evaluated using solely this single feature to determine how accurately a model could separate the two datasets based strictly on high-frequency energy.

*b) Control check:* To verify that successful timbral classification does not rely on this high-frequency shortcut, an 8 kHz low-pass filter was applied to all audio files across both datasets. This completely removed all acoustic information above 8 kHz. All timbral features were then re-extracted from the filtered audio, and the models were retrained and evaluated using the same grouped cross-validation setup to test whether they can still distinguish AdP from conventional rock when all audio frequencies above 8 kHz are removed.

## IV. Results

### A. Hypothesis Testing: H1 and H2

The hypotheses H1 and H2 for Angine de Poitrine (AdP) were evaluated using two-sided Mann–Whitney $U$ tests and Cliff's delta ($\delta$) effect sizes.

Table I reports the sample sizes, track medians, effect sizes, and $p$-values.

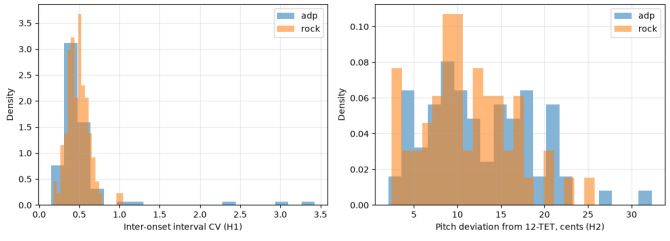



Fig. 1. Empirical density distributions for the hypothesis tests: coefficient of variation (CV) of inter-onset intervals (IOIs) (H1, left) and frame-level pitch deviation from 12-tone equal temperament (12-TET) in cents (H2, right).

*1) Rhythmic irregularity (H1):* The track-level median is 0.418 for AdP and 0.481 for GTZAN rock. Because the $p$-value ($p = 0.160$) exceeds the significance level ($\alpha = 0.05$), the test fails to reject the null hypothesis of equal distributions. Consequently, H1 cannot be supported. Cliff's delta ($\delta = -0.25$) confirms that the observed difference is small and points in the direction opposite to the hypothesis. As visualized in Fig. 1 (left), the distributions heavily overlap, with GTZAN displaying slightly wider dispersion.

TABLE I
Track-Level Hypothesis Tests for H1 and H2

| Feature | $N_{\text{AdP}}$ | $N_{\text{Rock}}$ | $\text{Median}_{\text{AdP}}$ | $\text{Median}_{\text{Rock}}$ | $\delta$ | $p$-value |
|---|---|---|---|---|---|---|
| rhy_ioi_cv | 12 | 100 | 0.418 | 0.481 | -0.25 | 0.160 |
| pit_yin_dev_mean | 11 | 56 | 11.750 | 10.359 | 0.13 | 0.504 |

*2) Pitch and microtonality (H2):* For H2, pitch was evaluated by testing microtonal deviation from standard Western 12-TET tuning. AdP displays a marginally higher median deviation (11.750 cents) than rock (10.359 cents). However, with $p = 0.504$, the test fails to reject the null hypothesis. Cliff's delta is negligible ($\delta = 0.13 < 0.147$ [8]), confirming that there is no meaningful separation between the two styles on this pitch feature. As shown in Fig. 1, the distributions for both groups overlap almost completely. Furthermore, reliable fundamental frequency estimates could only be obtained for 11 AdP tracks and 56 rock tracks, reflecting substantial data loss in polyphonic rock audio.

Taken together, failing to reject the null hypothesis means that these individual tests cannot prove that the two styles differ on onset irregularity or pitch deviation alone. Cliff's delta shows that the underlying differences are too small or negligible to establish separation.

### B. Classification

Supervised classification using logistic regression and a Support Vector Machine (SVM) with a radial basis function (RBF) kernel evaluated whether combining acoustic dimensions enables the models to identify AdP, using five-fold stratified grouped cross-validation.

*1) Primary rhythm and pitch classification (H3):* Hypothesis H3 was evaluated first using exclusively the rhythm and pitch feature families, directly testing whether the musical dimensions motivated by the research question contain sufficient information to identify AdP. Table II summarizes performance against the majority-class baseline.

TABLE II
Classification Performance Using Rhythm and Pitch Features (H3)

| Model | Accuracy | Precision | Recall | $F_1$ ($\sigma$) |
|---|---|---|---|---|
| Majority Baseline | 0.515 | 0.000 | 0.000 | 0.000 (0.000) |
| Logistic Regression | 0.873 | 0.858 | 0.888 | 0.870 (0.022) |
| SVM (RBF Kernel) | **0.904** | **0.880** | **0.929** | **0.903 (0.029)** |

Both models clearly outperform the majority baseline, which obtains 51.5% accuracy but $F_1 = 0$ because it predicts only the majority rock class. Using rhythm + pitch, logistic regression reaches 87.3% accuracy and $F_1 = 0.870$, while the RBF SVM reaches 90.4% accuracy and $F_1 = 0.903$. The SVM also achieves high AdP recall (0.929), meaning that most

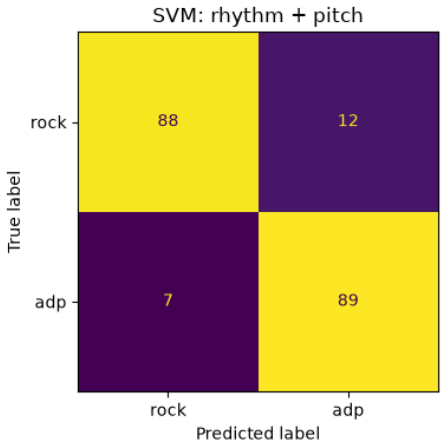



Fig. 2. Confusion matrix for the rhythm and pitch SVM under five-fold grouped cross-validation.

AdP excerpts are recognized correctly. This supports H3 even though H1 and H2 are not individually significant.

As shown in Fig. 2, across the grouped cross-validation predictions, the SVM correctly classifies 88 of 100 rock excerpts and 89 of 96 AdP excerpts. It produces 12 rock-to-AdP errors (false positives) and 7 AdP-to-rock errors (false negatives). The relatively balanced error pattern shows that the high $F_1$-score is not produced by favoring only one class.

*2) Feature family ablation:* Evaluating feature families separately reveals a clear numerical hierarchy, as reported in Table III and illustrated in Fig. 3. Rhythm alone yields the lowest performance ($F_1 = 0.711$ for logistic regression, 0.715 for SVM), while pitch alone achieves $F_1 = 0.840$ and 0.841, respectively. Combining rhythm and pitch increases $F_1$ to 0.870 for logistic regression and 0.903 for the SVM. Timbre alone yields the highest classification performance of any single feature family: logistic regression reaches an $F_1$-score of 0.977, and the SVM reaches 0.958. Evaluating all retained feature families combined yields $F_1 = 0.952$ for logistic regression and 0.967 for the SVM.

TABLE III
FEATURE-FAMILY ABLATION PERFORMANCE ($F_1$-SCORE ± STANDARD DEVIATION)

| Feature Representation | Logistic Regression | SVM (RBF Kernel) |
|---|---|---|
| Rhythm Only | 0.711 (0.080) | 0.715 (0.119) |
| Pitch Only | 0.840 (0.099) | 0.841 (0.070) |
| Rhythm + Pitch | 0.870 (0.022) | 0.903 (0.029) |
| Timbre Only | **0.977 (0.020)** | 0.958 (0.016) |
| All Features | 0.952 (0.046) | **0.967 (0.020)** |

*3) Recording-bandwidth confound check:* Table IV reports the classification results for the high-frequency diagnostic feature and the low-pass filtered timbre representations.

A measurable high-frequency difference exists between the datasets. The median energy above 9 kHz relative to total signal energy is $-30.6$ dB for AdP and $-38.5$ dB for GTZAN rock, representing a difference of 7.9 dB. Training classifiers solely on this single diagnostic feature yields an $F_1$-score of 0.643 for logistic regression and 0.843 for the nonlinear SVM,

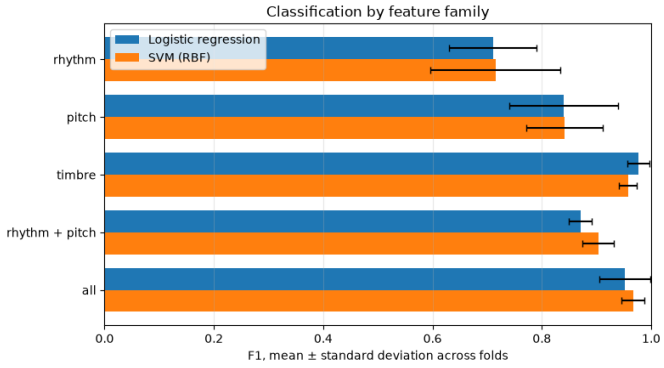



Fig. 3. Cross-validated $F_1$-score across isolated and combined feature families for Logistic regression and RBF SVM. Error bars denote $\pm 1$ standard deviation across the five validation folds.

confirming that high-frequency content alone carries class-predictive information.

Removing this frequency region, however, does not eliminate timbral separation. With the original full-band timbre features, $F_1$ is 0.977 for logistic regression and 0.958 for the SVM. After both classes are low-pass filtered at 8 kHz to eliminate all acoustic content above the cutoff, classification performance remains at $F_1 = 0.960$ for logistic regression and $F_1 = 0.974$ for the SVM.

TABLE IV
BANDWIDTH CONFOUND DIAGNOSTIC AND LOW-PASS FILTER COMPARISON

| Feature Set | Logistic Regression $F_1$ | SVM (RBF) $F_1$ |
|---|---|---|
| Energy above 9 kHz | 0.643 (0.132) | 0.843 (0.071) |
| Timbre (Full Band) | 0.977 (0.020) | 0.958 (0.016) |
| Timbre (Low-Pass at 8 kHz) | 0.960 (0.024) | 0.974 (0.019) |

## V. DISCUSSION

### A. Distinguishing Musical Characteristics

As demonstrated by the tests in Table I, Angine de Poitrine (AdP) did not show statistically significant separation from GTZAN rock on the selected individual timing and tuning measures ($p > 0.05$ for both H1 and H2). Most notably, the test for rhythmic irregularity pointed in the direction opposite to the hypothesis: Cliff's delta was negative ($\delta = -0.25$), with AdP displaying lower median onset variability than rock (0.418 vs. 0.481, $p = 0.160$). A plausible explanation is that the coefficient of variation (CV) of inter-onset intervals (IOIs), $CV_{\mathrm{IOI}}$, measures only the time gap between consecutive onsets, rather than which beats are played louder. When unconventional rhythms are played with steady note timing, the time intervals remain uniform, resulting in a low $CV_{\mathrm{IOI}}$. In contrast, conventional rock frequently alternates between pauses and fast drum fills, creating uneven time gaps that increase its onset variability. The feature appears to have captured pauses and dynamic shifts rather than rhythmic complexity.

Similarly, the test for microtonal deviation showed no significant difference between the two genres. In fact, the pitch tracker failed to extract a stable fundamental frequency

from 44 of the 100 rock tracks due to dense, polyphonic instrumentation (drums, bass, and distorted guitars playing simultaneously). For the remaining tracks where pitch could be tracked, the algorithm sampled audio continuously in short time slices. In conventional rock, standard playing techniques, such as electric guitar string bends, naturally glide between standard notes. The algorithm counted each of these brief transitions as an off-grid pitch, which likely inflated the measured pitch deviation for rock music. Consequently, the feature could not distinguish expressive rock gestures from deliberate microtonal tuning in AdP.

### B. Classification Performance

While statistical tests showed no standalone separation, supervised classification revealed that distinguishing AdP from conventional rock is an easily solvable multivariate task. This demonstrates that the fingerprint is distributed across combinations of features rather than isolated in individual parameters.

The models performed differently depending on the feature representation:

*1) Rhythm and pitch:* The nonlinear radial basis function (RBF) support vector machine (SVM) outperformed logistic regression (0.903 versus 0.870 $F_1$), indicating that rhythmic pulse and harmonic chroma interact nonlinearly. A curved decision boundary was necessary to capture these multidimensional musical relationships.

*2) Timbre alone:* Logistic regression outperformed the RBF SVM (0.977 versus 0.958 $F_1$), achieving the highest score across all experiments. This shows that the timbral and mel-frequency cepstral coefficients (MFCCs) feature space is already linearly separable; introducing nonlinear kernel transformations adds unnecessary complexity.

### C. Importance of Different Feature Groups

As shown in Table III, the feature-family ablation revealed a consistent hierarchy of predictive power across both classifiers:

$$\text{Rhythm Only} < \text{Pitch Only} < \text{Rhythm+Pitch} < \text{Timbre Only}$$

Pitch features contributed substantially more discriminative power than rhythm alone for both models, indicating that differences in harmonic palette and chordal vocabulary provide stronger stylistic cues than onset timing. Crucially, timbral features overwhelmingly dominated classification. Although MFCCs lack the direct musicological interpretability of metric structures or tuning systems, they capture the holistic acoustic fingerprint of the music, including instrumentation, amplifier distortion, and vocal delivery. In fact, adding rhythm and pitch descriptors to the timbral representation did not improve performance, demonstrating that the overall sound texture carries all the information necessary to identify AdP.

### D. Limitations

Several methodological limitations qualify these findings.

First, the positive class comprises only twelve studio tracks from a single artist across two releases; consequently, the models learn the idiosyncratic acoustic signature of AdP rather than a generalized representation of unconventional rock.

Second, GTZAN is known to contain recording repetitions, varying fidelities, and dataset faults [9]. While the 8 kHz low-pass filter control confirmed that high-frequency recording bandwidth does not account for timbral separation, other acoustic differences below 8 kHz, such as instrumentation, guitar distortion, and modern studio production, still differ between the datasets. Consequently, the classifiers may capture a combination of musical composition, production style, and artist-specific sound.

Finally, monophonic pitch tracking proved unreliable on dense polyphonic rock mixes (failing on 44 rock tracks), which prevented direct microtonal tuning features from being evaluated in the classification models.

### E. Further Work

To address these limitations, several extensions are proposed for future research.

First, to overcome the single-artist constraint, future studies should evaluate a broader corpus comprising multiple artists with unconventional, progressive, or microtonal styles. Evaluating a multi-artist dataset would test whether machine learning models can learn generalized stylistic conventions rather than memorizing the idiosyncratic acoustic and vocal signature of a single band.

Second, to control for production and mastering differences between modern releases and historical datasets such as GTZAN, future work should benchmark unconventional styles against a contemporary rock corpus recorded under comparable modern studio standards. Evaluating recordings from similar production eras would isolate compositional characteristics from historical recording chains.

## VI. Conclusion

This study investigated whether the unusual musical style of Angine de Poitrine (AdP) can be distinguished from conventional rock using computational audio features, focusing specifically on rhythmic irregularity and microtonality. The hypotheses H1 and H2 were not supported: neither rhythmic irregularity nor microtonal pitch deviation showed a statistically significant difference from conventional rock, with the observed rhythmic trend even pointing in the direction opposite to the hypothesis. However, combining rhythm and pitch descriptors in a supervised classifier strongly supported H3, clearly outperforming the majority baseline. Under five-fold grouped cross-validation, logistic regression reached an $F_1$-score of 0.870, while the nonlinear radial basis function (RBF) support vector machine (SVM) reached 0.903, correctly classifying 89 of 96 AdP excerpts and 88 of 100 rock excerpts.

The feature-family comparison showed that timbre was by far the most discriminative representation ($F_1 = 0.977$), indicating that the primary measurable differences between AdP and conventional rock are driven by broader spectral characteristics, instrumentation, and vocal textures rather than rhythmic irregularity or microtonality alone. Furthermore,

while an empirical 7.9 dB high-frequency bandwidth difference was identified between the modern AdP digital releases and the GTZAN corpus, strong timbral classification persisted ($F_1 \geq 0.960$) after low-pass filtering both datasets at 8 kHz, demonstrating that the separation cannot be explained solely by this bandwidth difference. Overall, AdP possesses a clearly distinguishable audio profile relative to conventional rock, but the empirical results do not support the conclusion that this distinction is primarily caused by rhythmic irregularity or microtonality.

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